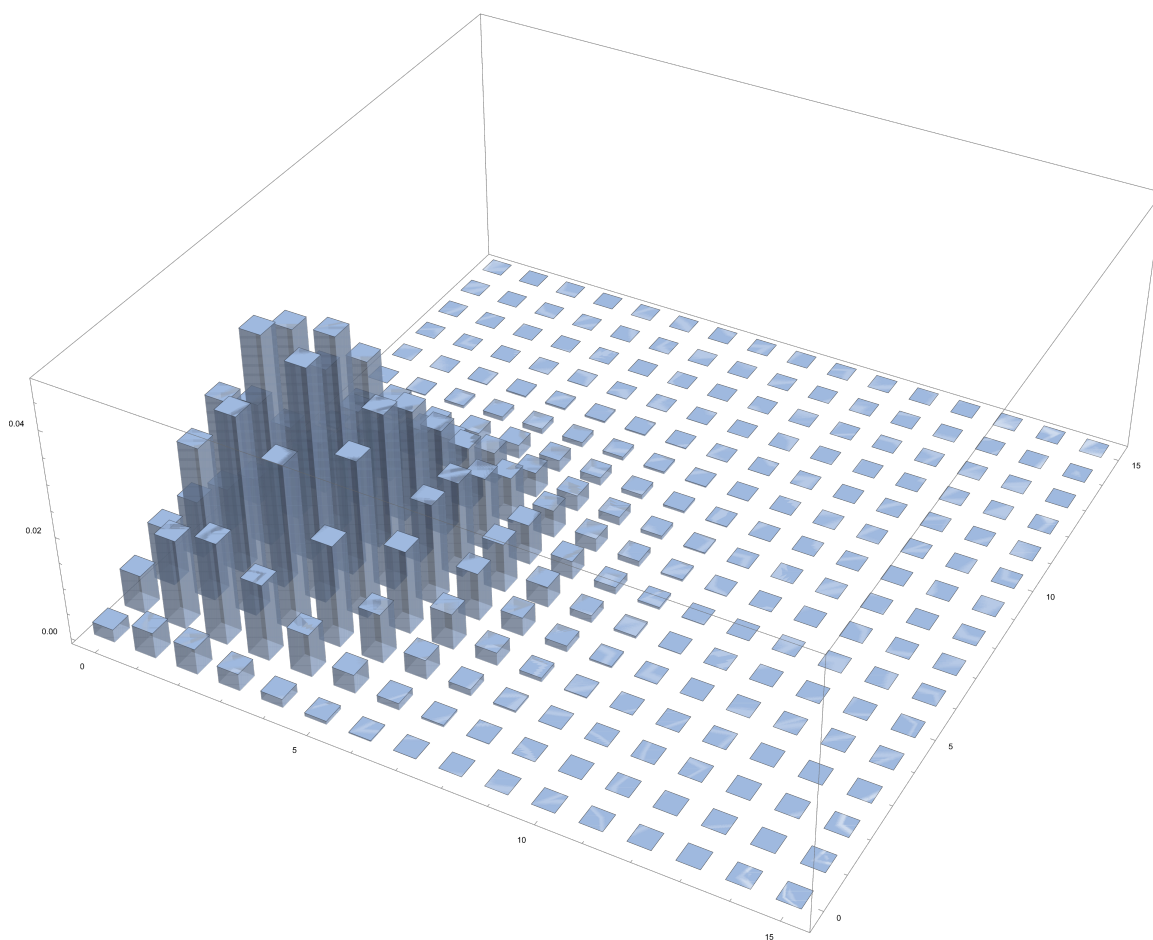




Probability and Combinatorics

The Stochastic Art of Throwing Dice

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May 24, 2026

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Introduction

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CHAPTER 1

Foundations of Probability Theory

1.1 Probability Axioms and Conceptual Foundations

1.2 Conditional Probability, Inclusion-Exclusion, and Independence

1.3 Combinatorics and Counting Principles

1.3.1 Permutations and Combinations

1.3.2 Stars and Bars

Weakly Descending

How many 4-digit numbers have their digits in weakly descending order? In these numbers, every digit beginning with the hundreds digit is less than or equal to the digit before it.

SOLUTION:

Consider the following scenario. We have, in a row, 9 boxes with labels in descending order from 9 to 1. We also have 4 identical balls. It is key to realize that the number of combinations of placing the balls in the boxes is synonymous with the original question. If each ball is a digit and corresponds to the digit of the box it is in, then that combination is a unit ordering. For example,

$$d_1|||d_2d_3||d_4|||$$

corresponds to 9553. The only arrangement that yields an invalid solution is $|||||||d_1d_2d_3d_4$, which corresponds to 0000. Thus, the number of valid combinations is yielded by stars and bars:

$$\binom{9+4-1}{4} - 1 = \frac{13(12)(11)(10)}{(4)(3)(2)} - 1 = \boxed{714}$$

1.3.3 Symmetry and Overcounting

Firm Circular Seating

Six people sit around in a circular table. Romeo and Juliet wish to sit together. Julius does not wish to sit next to Brutus. Rosencrantz and Guildenstern do not care where they sit. In how many distinguishable ways can they be seated? Two seating arrangements are distinguishable if one cannot be rotated parallel to the floor to form the other. (2017 17)

CHAPTER 2

Random Variables

2.1 Common Probability Distributions

- 2.1.1 Discrete Uniform
- 2.1.2 Continuous Uniform
- 2.1.3 Binomial Distribution
- 2.1.4 Geometric Distribution
- 2.1.5 Negative Binomial Distribution
- 2.1.6 Hypergeometric Distribution
- 2.1.7 Normal Distribution

2.2 Expectation and Variance

2.3 Transformations of Random Variables

2.4 Limit Theorems

- 2.4.1 Law of Large Numbers
- 2.4.2 Central Limit Theorem
- 2.4.3 Empirical Rule

2.5 Markov's and Chebyshev's Inequalities

2.6 Uncommon Probability Distributions

- 2.6.1 Poisson Distribution
- 2.6.2 Negative Hypergeometric Distribution
- 2.6.3 Chi-Squared Distribution
- 2.6.4 Exponential Distribution
- 2.6.5 Dirac Delta Function

CHAPTER 3

Advanced Expectation

3.1 Linearity of Expectation

3.2 Indicator Variables

Roundtable Handshakes

100 people sit around a roundtable. Each person raises either their left hand, right hand, or both hands, each with $1/3$ probability. If two adjacent people raise their adjacent hands, they shake hands (e.g., a person who raises their right hand and a person to their right raises their left hand would shake hands). What is the expected number of handshakes?

SOLUTION:

Let H_i be the event that person i shakes hands with person $i + 1$ (with person 100 shaking hands with person 1). Then the total number of handshakes is

$$H = \sum_{i=1}^{100} H_i$$

By linearity of expectation,

$$E(H) = \sum_{i=1}^{100} E(H_i)$$

Now we analyze H_i . The probability person i raises their right hand is $1/3 + 1/3 = 2/3$. The probability person $i + 1$ raises their left hand is $1/3 + 1/3 = 2/3$. Thus, the probability that person i and person $i + 1$ shake hands is $P(H_i) = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$. Therefore,

$$E(H) = \sum_{i=1}^{100} \frac{4}{9} = \boxed{\frac{400}{9}}$$

3.3 The Principle of Symmetry

First Ace

Cards are drawn from the top of a shuffled regular 52-card deck. What is the expected number of cards needed to draw the first ace?

SOLUTION:

Third Highest Card

From the set of the first 24 positive integers, six are randomly and uniformly chosen without replacement. What is the expected value of the third highest card?

SOLUTION:

We can use a stars-and-bars method of thinking to solve this problem.

3.4 Coupon Collector Problem

3.5 Markov Chains

Rabbit Jump 'Til Termination

A rabbit starts at the origin on a number line. At each second, it jumps to the right with probability $1/3$ and to the left with probability $2/3$. The rabbit stops when it reaches -2 , 2 , or after 8 jumps, whichever comes first. What is the expected number of jumps?

SOLUTION 1:

Note that the jumps must terminate on an even second. With this in mind, we can calculate the probability it terminates at each even second. Denote e_n as the probability it terminates on the n^{th} second.

$$e_2: \text{LL or RR, so } e_2 = \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = \frac{5}{9}$$

$$e_4: \text{LR or RL, then probability it terminates in two seconds, or } e_2. \text{ Thus } e_4 = 2\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)e_2 = \frac{20}{81}.$$

$$e_6: \text{We follow the same logic as above. } e_6 = 2\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)e_4 = \frac{80}{729}$$

$$e_8: \text{We must include ending at 0 in addition to -2 and 2.}$$

$$e_8 = 2\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)e_6 + \left(2\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)\right)^4 = \frac{64}{729}$$

Thus,

$$E(X) = 2 \left(\frac{5}{9} \right) + 4 \left(\frac{20}{81} \right) + 6 \left(\frac{80}{729} \right) + 8 \left(\frac{64}{729} \right) = \boxed{\frac{2522}{729}}$$

SOLUTION 2:

We can use a “survival” probability to determine the expected value. Let e_n be the probability of a given rabbit not being terminated at time n . Again note that the rabbit can only terminate on an even second. Then,
 $e_1 = 1$ $e_2 = \frac{4}{9}$ $e_3 = \frac{4}{9}$ $e_4 = \frac{16}{81}$ $e_5 = \frac{16}{81}$ $e_6 = \frac{64}{729}$ $e_7 = \frac{64}{729}$ $e_8 = \frac{64}{729}$ Thus,

$$E(X) = \sum_{n=1}^8 e_n = 1 + 1 + \frac{4}{9} + \frac{4}{9} + \frac{16}{81} + \frac{16}{81} + \frac{64}{729} + \frac{64}{729} = \boxed{\frac{2522}{729}}$$

3.6 Runs and Sequences

3.7 Random Walks

Karthik is on a vertex of a pentagon. Every minute, he randomly moves to an adjacent vertex. What is the expected number of minutes until he returns to his original vertex.

SOLUTION 1 (Markov Chain):

SOLUTION 2 (Utilising Random Walks):

Again, due to symmetry, where we begin doesn't matter. However, the key insight is realizing that after the first step, we now have a random walk with bounds at -1 and 4 . The expected time for the run to terminate is $ab = (1)(4) = 4$. Thus, the expected time is $1 + 4 = 5$.

3.7.1 Gambler's Ruin

3.8 Advanced Geometric Thinking

Ten points are uniformly, independently, randomly chosen on the interval $[0, 1]$. What is the expected minimum distance between any two points?

Three points are independently and randomly placed on the unit circle. What is the expected arc length of the arc containing the point $(1, 0)$?

Points are uniformly, independently, and randomly chosen on the interval $[0, 1]$. What is the expected number of points needed to be generated so that the sum exceeds one?

3.9 Advanced Probabilistic Thinking

Painting the Circumcircle

A circle is drawn. A first point is randomly placed on the circle, and then a second. Draw a green line clockwise from the first point to the second point. A third point is randomly placed on the circle, then a fourth. Draw a red line counterclockwise from the third point to the fourth point. What is the probability that the two line segments intersect?

SOLUTION:

Acute Inscribed Triangle

Three points are independently and randomly placed in a circle, and connected to form a triangle. What is the probability that the triangle is acute?

3.9.1 Reduced Spaces

2 Dice before 2 Dice

Roll a die until a 2 and a 3 are rolled at least once. What is the probability that exactly 2 6s appear?

SOLUTION:

3.10 Combinations with Expectation

Draw cards repeatedly from a standard 52-card deck without replacement. What is the expected number of cards needed to draw until you get an ace?

SOLUTION:

Consider a three-by-six grid. Place the numbers 1 through 18 into the grid, uniformly at random without replacement. What is the expected value of the smallest number in the second row?

3.11 Game Theory and Nim

CHAPTER 4

Advanced Counting Techniques

4.1 Derangements

A derangement is a permutation of a set of elements where no object appears in its original position. Say, for example, that three people, A, B, C, put their coat on a coat rack. Later, those three people randomly take a coat when leaving. A possible derangement is that A grabs C's coat, B grab A's coat, and C grabs B's coat.

The number of derangements of a set of size n is known as the n^{th} derangement, commonly denoted as $!n$, $D(n)$, or D_n . There are several helpful formulas for derangements:

$$D(n) = (n-1) (D(n-1) + D(n-2))$$

$$!n = n! \sum_{r=0}^n \frac{(-1)^r}{r!}$$

$$!n = \left\lfloor \frac{n! + 1}{e} \right\rfloor$$

$$\lim_{n \rightarrow \infty} \frac{!n}{n!} = \frac{1}{e}$$

4.2 Vandermonde's Identity

Consider a 9-by-9 grid with a point on the bottom left corner and another point on the top right corner. Each second, the bottom left point can move either up or right one space, and the top right point can either move down or left one space. In how many possible ways can the points meet?

4.3 Burnside's Lemma

4.4 Distinguishability

4.5 Catalan Numbers

4.6 Generating Functions

4.7 Pigeonhole Principle

4.8 Partitions

4.9 Recurrence Relations

Appendices

APPENDIX A

Title

There once was a very smart but sadly blind duck. When it was still a small duckling it was renowned for its good vision. But sadly as the duck grew older it caught a sickness which caused its eyesight to worsen. It became so bad, that the duck couldn't read the notes it once took containing much of inline math just like its favoured equation: $d = u_c \cdot k$. Only displayed equations remained legible so it could still read

$$d = ra^ke.$$

That annoyed the smart duck, as it wasn't able to do its research any longer. It called for its underduckling and said: "Go, find me the best eye ducktor there is. He shall heal me from my disease!"

